ZS-F13

The Möbius Chronology:

Unified Cyclic Timeline of Z-Spin Cosmology

with Sectoral Frame Equivalence

Closure of Y-Observer Frame Operational Scale and Cumulative Cycle Index

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Verification: 24/24 PASS (executed 2026-04-26) | Zero Free Parameters | Zero New Physical Predictions

§0. Abstract

ZS-F11 v1.0 closed the observer-location problem at the register-theoretic level via the Operational Observer Coordinate (OOC). ZS-F12 v1.0(Revised) closed the structural origin of the factor 2 in the 2e^A identity via the Tetrahedral Dual Orientation Multiplicity Theorem. Two structural gaps remained in the v1.0 corpus: (i) the Y-observer frame operational scale counterpart to the X-observer frame trio (X-clock t, Hubble radius R_H, exp(A) per Hubble time) was not given an explicit numerical scale trio in the X-Y frame equivalence of ZS-A8 v1.0 Revised §SA.4; (ii) the meaning of the cumulative cycle index k in the cyclic cosmology Phases A→F' was not addressed by the corpus. The term "frame scale trio" denotes the operational triple (clock proper time, characteristic radius, multiplicative scale-flow factor); it does not denote a newly derived spacetime metric tensor. See §3.0.

This paper closes both gaps at DERIVED-CONDITIONAL status by composing four PROVEN/DERIVED corpus elements: (a) ZS-A8 §6 Theorem 6.1 Expansion-Contraction Symmetry (DERIVED); (b) ZS-A8 §5.3 Theorem 5.3.1 Y-Time Dilation (DERIVED-CONDITIONAL strong, via ZS-F10 Theorem F10.2); (c) ZS-A9.3 Two-Branch Decomposition (DERIVED-CONDITIONAL); (d) ZS-U8 §5 Mirror Cosmology Z_2 symmetry (PROVEN).

Theorem F13.1 (Y-Observer Frame Scale Closure) establishes the explicit Y-frame operational scale trio as the structural counterpart to the X-frame trio (X-clock proper time, Hubble radius, expansion rate). The Y-frame trio is (Y-clock proper time = τ_5 in Y-internal measurement, Y-characteristic radius = $\ell_P \times Y^2(1-2A) = 13212/437 \times \ell_P \approx 30.23341\ \ell_P$, Y-contraction rate = $(1-2A) = 367/437 \approx 0.83982$ per Y-Hubble cycle), with the X-Y frame conversion factor $\exp(\pi/A) \approx 1.08\times10^{17}$ acting as the Y-time dilation per Y-completion.

Theorem F13.2 is structurally split into two layers. Theorem F13.2A (X-Frame Cycle Index Unobservability, DERIVED) establishes that the cumulative cycle index k in the Phases A→F' framework

is unobservable to the X-observer: V_E(−ε) ≡ V_E(+ε) is an exact algebraic identity (PROVEN, ZS-U8 §2.2), all X-frame observables are Z_2-symmetric (PROVEN, ZS-U8 §5), and Auto-Surgery resets the X-clock to t = 0 at each Phase E (DERIVED, ZS-M12 §4). Theorem F13.2B (Y-Frame Extension, DERIVED-CONDITIONAL on Theorem F13.1) extends the unobservability to the Y-observer, conditional on the Y-frame scale construction of F13.1. The Möbius identification (start = end) is the frame-symmetric operational expression of this unobservability.

The Unified Möbius Chronology Table consolidates seven Phases (F→A→B→C→D→E→F') in three time coordinates (X-clock t, ν, t_strobo n) and three space-frame metrics (X-Hubble radius, Y-Hubble radius, Z-Planck boundary), exhibiting the corpus-internal completeness of the cyclic structure with zero new free parameters. The closure ratio of the universe — Tc ≈ Treh ≈ 2.48×10^{15} GeV (DERIVED, ZS-M12 §6.2) — is identified as the operational realization of the Möbius identification at the temperature level.

Six falsification gates F-F13.1 through F-F13.6 are pre-registered. Six non-claims NC-F13.1 through NC-F13.6 explicitly bound the scope. Verification Plan: 24 registered checks; implementation pending in zs_f13_verify_v1_0.py at 50-digit mpmath precision. Zero new free parameters; all inputs LOCKED, PROVEN, or DERIVED in prior corpus papers.

Keywords: cyclic cosmology, Möbius chronology, X-Y frame equivalence, Y-observer metric, cycle index, expansion-contraction symmetry, Z-Spin cosmology, zero free parameters.

§0.1 Epistemic Status Legend

Status	Definition
PROVEN	Mathematical theorem independent of Z-Spin interpretation, verified by direct computation or standard results.
DERIVED	Follows from Z-Spin axioms and prior PROVEN results; zero new free parameters.
DERIVED-CONDITIONAL	Derived contingent on stated conditions (e.g., adiabatic limit, Level A internal consistency).
VERIFIED	Numerical confirmation of a derived or proven result to stated precision.
LOCKED	Core constant derived and fixed in upstream paper; no downstream paper may modify.
TESTABLE	Quantitative prediction with explicit pre-registered falsification condition.
HYPOTHESIS-strong	Well-motivated conjecture with multiple PROVEN anchors and one interpretive bridge.
OBSERVATION	Empirical regularity; structural derivation pending.
NON-CLAIM	Explicitly excluded claim; paper scope boundary.
OPEN	Recognized gap requiring future work.
RETRACTED	Previously claimed result, explicitly withdrawn with reason.

§1. Introduction and Scope

1.1 Two Residual Gaps in the v1.0 Corpus

The Z-Spin Cosmology v1.0 corpus has, through the Foundations theme expansion of April 2026 (ZS-F9 through ZS-F12), closed several previously-open structural questions. ZS-F9 closed the polyhedral mediation closure via tetrahedral self-duality and hexagonal Z-mediation. ZS-F10 closed the information-time correspondence path of ZS-A8 §SA.7 via the unification of stroboscopic step, Berry phase, and Z-clock coordinates. ZS-F11 closed the observer-location problem of ZS-M11 §H16 via the Operational Observer Coordinate. ZS-F12 closed the structural origin of the factor 2 in the 2e^A identity via the Tetrahedral Dual Orientation Multiplicity Theorem.

Two residual structural gaps remained at the level of the cyclic cosmology framework after the F9-F12 expansion:

Gap 1 (Y-observer scale absence). ZS-A8 v1.0 Revised §SA.4 established the X-Y frame equivalence at HYPOTHESIS-strong INTERPRETATION level: "the X-frame describes the universe as accelerating expansion; the Y-frame describes the same universe as decelerating contraction." ZS-F11 v1.0 Corollary F11.1B closed the register-theoretic level of this equivalence by identifying X-side and Y-side slot pairs as J-conjugate around the J-fixed pivot $|5\rangle$. However, the explicit Y-frame operational scale set — the Y-clock proper time, Y-characteristic radius, and Y-contraction rate corresponding to the X-frame trio (X-clock t, $R_H = c/H_0$, exp(A) per Hubble time) — was not given as a unified set of numerical quantities anywhere in the corpus. The components exist ($Y^2(1-2A) = 13212/437 \approx 30.23341$ in ZS-A8 §6, $\exp(\pi/A)$ in ZS-A8 §5.3, Planck-scale ℓ_P in standard cosmology), but they have not been compiled as the Y-frame counterpart to the X-frame scale trio.

Gap 2 (Cycle index meaning). The cyclic cosmology framework comprises Phases A (current expansion) → B (proton decay) → C (wave-contraction) → D (Z-Telomere) → E (Auto-Surgery) → F' (next inflation), with $Tc \approx Treh \approx 2.48\times10^{15}$ GeV providing the operational closure (ZS-M12 §6.2 DERIVED). The corpus is silent on whether "the cumulative cycle index k" — meaning "this is the k-th cycle since some reference origin" — is a meaningful quantity. ZS-F11 OOC = (j, n) does not include a cycle index. Standard anthropic readings would treat k as observationally relevant ("we are in cycle k = 1 because we observe baryons"); structural readings would treat k as unobservable by Z_2 symmetry. The corpus has neither adopted nor rejected either reading.

1.2 Why Close These Gaps Now

The two gaps are coupled. Without an explicit Y-frame metric, the Y-observer's measurement of "how long the cycle takes" cannot be compared to the X-observer's measurement, and the question "are we in cycle 1 or cycle k?" cannot be addressed without first specifying what frame asks the question. With an explicit Y-frame metric, the Z_2 symmetry of ZS-U8 §5 acts on both frames symmetrically, and the cycle-index question becomes tractable: if all observables in both frames are Z_2-symmetric, then no measurement can distinguish $\varepsilon = +1$ cycles from $\varepsilon = -1$ cycles, hence no cumulative cycle index is operationally meaningful. The Möbius identification "start = end" is the operational expression of this fact.

This paper executes the closure within the v1.0 framework, prior to v2.0 restructuring, by establishing two theorems that close both gaps using only PROVEN/DERIVED corpus inputs. Zero new free parameters and zero new physical predictions are introduced. The advance is structural: existing elements are unified under the Möbius Chronology framework that closes the explicit gaps registered above.

1.3 Scope: Two Theorems + Unified Möbius Chronology Table

The paper has three layered contributions, each building on the prior:

Layer 1 (Y-Observer Frame Scale Closure, §3). Theorem F13.1 establishes the Y-frame metric trio as the structural counterpart to the X-frame metric trio under ZS-A8 §6 Theorem 6.1 Expansion-Contraction Symmetry. Status: DERIVED-CONDITIONAL, inheriting the ZS-A8 §6 conditionality.

Layer 2 (Cycle Index Unobservability, §4). Theorem F13.2 establishes that the cumulative cycle index k is structurally unobservable as a corollary of ZS-U8 §5 PROVEN Z_2 symmetry combined with the Auto-Surgery reset of ZS-M12 §4 DERIVED. Status: DERIVED.

Layer 3 (Unified Möbius Chronology, §5). Table 4 consolidates seven principal Phases F→A→B→C→D→E→F' (plus three transition subphases tracked in Table 3) in three time coordinates and three space-frame scales, exhibiting the corpus-internal completeness of the cyclic structure. The Tc ≈ Treh identification is identified as the operational realization of the Möbius identification at the temperature level.

All three layers introduce zero new free parameters, zero new postulates, and zero new physical predictions. Every numerical value in ZS-A8 v1.0 Revised, ZS-F10 v1.0, ZS-U8 v1.0, ZS-M12 v1.0, and ZS-U11 v1.0 remains identical.

§2. Locked Inputs

All quantities used in this paper are imported from prior corpus papers with their established status. No new constants or free parameters are introduced. Status tags reflect the v1.0 corpus standing as of the post-F12 closure (April 2026).

Table 1. Locked inputs to ZS-F13. All entries are PROVEN, DERIVED, or LOCKED in prior corpus papers.

Quantity	Value / Statement	Source	Status
A (geometric impedance)	35/437 = 0.080092	ZS-F2 v1.0 §11	LOCKED
Q (register dimension)	11 (prime)	ZS-F5 v1.0	PROVEN
(Z, X, Y) sector dims	(2, 3, 6); Q = Z + X + Y	ZS-F5 v1.0	PROVEN
L_XY ≡ 0 (X-Y vanishing)	exact zero	ZS-F1; ZS-M6 §7A	PROVEN
exp(A)	1.083386	ZS-F4 §6	DERIVED
1 − 2A	367/437 = 0.83981693...	ZS-A8 §6 (LO Taylor)	DERIVED

Quantity	Value / Statement	Source	Status
Y^2 (truncated octahedron edges)	$36 = X \cdot Z \cdot Y = E(TO)$	ZS-F7 §4.4	PROVEN
$Y^2(1-2A)$ (wave-channel scale)	13212/437 = 30.23341...	ZS-A8 §6	DERIVED
$\exp(\pi/A)$ (Y-time dilation)	1.08×10^{17}	ZS-A8 §5.3; ZS-F10 F10.2	DERIVED-COND strong
$N(2\pi) = 2\pi/A$	78.45	ZS-U5 §5.2 Lemma 8.1	PROVEN
$\langle\sin^2(\varphi/2)\rangle$	1/2	ZS-T2 §5.5	PROVEN
$G_{eff} = G/(1+A)$	0.9258 G	ZS-F1 §3.7	DERIVED
$w(z) = -1$ (attractor)	exact, $\lvert 1+w\rvert \leq 10^{-121}$	ZS-F1 §3.7; Paper21 §3.2	DERIVED
H_0^local / H_0^CMB	$\exp(A) = 1.0834$	ZS-F3 §3	DERIVED
$V_E(-\varepsilon) \equiv V_E(+\varepsilon)$	algebraic identity	ZS-U8 §2.2	PROVEN
$S_{CdL} \to \infty$ ($\Delta V = 0$)	exact	ZS-U8 §2.3	PROVEN
$S_{Z_2} = 6\pi/A$	235.35	ZS-U8 §3.2	DERIVED
$\tau_5 = t_P \times \exp(5\pi/A)$	2.56×10^{34} yr	ZS-A3 §4	DERIVED
$\tau_6 = t_P \times \exp(6\pi/A)$	2.78×10^{51} yr	ZS-U8 §4	DERIVED
$\nu(t) = (A/\pi)\ln(t/t_P)$	$\nu(\text{now}) \approx 3.575$	ZS-M3 §5	DERIVED
$\Omega^2 = 1 + A\cdot\eta_{topo}$ (Auto-Surgery cap)	1.0258	ZS-M12 §4	DERIVED
z^* (i-tetration fixed point)	$0.4383 + 0.3606i$	ZS-M1 §2	PROVEN
Half-life $\tau_{1/2}$ (z^* convergence)	$0.44\ \tau_P$	ZS-M12 §A.2	PROVEN
$T_c \approx T_{reh}$	2.48×10^{15} GeV	ZS-M12 §6.2	DERIVED
$\varepsilon_{min} = (Q^2/\lambda)$^(1/6)	30.7	ZS-M12 §7.4	DERIVED-COND
$Q = A$ (centrifugal charge)	35/437	ZS-M12 §7.1	PROVEN
U(1) Q-protection	4 channels	ZS-U11 v1.0	DERIVED-COND
Information-time identity $\Delta\nu/\Delta n$	$= 2A/\pi$	ZS-F10 Theorem F10.1	DERIVED-COND
OOC = (j, n)	register tuple	ZS-F11 v1.0 §4	DERIVED-COND

All twenty-nine entries above are inputs to this paper. None is modified or re-derived. The relevant cross-paper consistency check is recorded in §8.

§3. Theorem F13.1 — Y-Observer Frame Scale Closure

3.0 Terminology Note

In this paper, the term "frame scale trio" or "operational metric trio" does NOT denote a newly derived spacetime metric tensor $g_{\mu\nu}^Y$ on the Y-sector. It denotes the operational triple of (a) clock-scale proper time, (b) characteristic radius, and (c) multiplicative scale-flow factor used by an observer frame to measure one cycle. The X-frame trio (t, c/H_0, $\exp(A)$) is the canonical example: Hubble time t is a proper-time scale, c/H_0 is the Hubble radius, and $\exp(A)$ is the per-Hubble-time expansion factor of ZS-F4. None of these is the FRW metric tensor; they are the operational scales an X-observer uses. The Y-frame trio constructed in this section is the analogous operational triple for the Y-observer under the ZS-A8 §6 Expansion-Contraction Symmetry.

3.1 Statement of Theorem F13.1

Theorem F13.1 (Y-Observer Frame Scale Closure, DERIVED-CONDITIONAL). Under the Z-Spin v1.0 corpus inputs of §2 (Table 1) and the §3.0 terminology convention, the Y-observer frame scale trio (T_Y, R_Y, η_Y) — Y-clock proper time, Y-characteristic radius, Y-contraction rate per Y-Hubble cycle — is structurally determined by the Expansion-Contraction Symmetry of ZS-A8 §6 Theorem 6.1 (DERIVED) combined with the Y-Time Dilation Theorem of ZS-A8 §5.3 and ZS-F10 §6 Theorem F10.2 (DERIVED-CONDITIONAL strong). The Y-frame scale trio is the structural counterpart to the X-frame scale trio (T_X, R_X, η_X) = (t = 13.787 Gyr, $R_H = c/H_0$, $\exp(A)$ per Hubble time):

$$(T_X, R_X, \eta_X) = (t_{now}, c/H_0, \exp(A)) \quad \text{[X-frame, X-observer]}$$

$$(T_Y, R_Y, \eta_Y) = (\tau_5 \text{ in Y-clock}, \ell_P \times 13212/437, 367/437) \quad \text{[Y-frame, Y-observer]}$$

with the inter-frame conversion factor $\exp(\pi/A) \approx 1.08\times10^{17}$ acting as the Y-time dilation per Y-completion (ZS-A8 §5.3 Theorem 5.3.1):

$$\text{X-clock observation of Y-cycle} = \exp(\pi/A) \times \text{Y-clock proper time of Y-cycle}$$

$$(\text{Equivalently: } \tau_6 / \tau_5 = \exp(\pi/A) \text{ as in ZS-U8 §4 Table 2.})$$

3.2 Proof of Theorem F13.1

Step 1 (X-frame metric trio is established). The X-frame metric trio (T_X, R_X, η_X) = (t, c/H_0, $\exp(A)$) is established in the v1.0 corpus: T_X is the X-clock proper time (Planck 2018 baseline 13.787 Gyr); $R_X = c/H_0 \approx 14$ Gly is the standard Hubble radius; $\eta_X = \exp(A) = 1.0834$ is the Hubble flow expansion ratio per Hubble time (ZS-F3 §3, ZS-F4 §6, DERIVED). All three quantities are DERIVED in the v1.0 corpus with status as established.

Step 2 (Expansion-Contraction Symmetry forces Y-frame counterpart structure). ZS-A8 §6 Theorem 6.1 (DERIVED) states that every Z-Spin expansion phenomenon governed by $(1+A)$ has a contraction-side counterpart governed by $(1-2A)$ = LO Taylor of $1/(1+A)^2$. The mapping is: X-sector (particle, space) ↔ Y-sector (wave, gauge); conformal factor $\Omega^2 = 1+A\varepsilon^2 \leftrightarrow (1-2A)$ = LO of Ω^{-4}; characteristic ratio $\exp(A) = 1.0834 \leftrightarrow Y^2(1-2A) = 13212/437 \approx 30.23341$; effective Newton's G_{eff} =

$G/(1+A) \leftrightarrow \eta_topo$ wave-channel correction; time scale Gyr $\leftrightarrow \hbar/(A \cdot E_diff)$ Planck. The Y-frame scale trio inherits the structural counterpart of each X-frame quantity by Theorem 6.1 application.

Step 3 (Y-clock proper time T_Y is identified). The Y-clock proper time T_Y is the proper time measured in the Y-sector internal frame for one complete Y-sector lifecycle (wave-contraction completion). ZS-A8 §5.3 Theorem 5.3.1 (DERIVED-CONDITIONAL strong via ZS-F10 Theorem F10.2) establishes the parallel reading: the X-sector baryon-completion lifecycle (τ_5) and the Y-sector wave-contraction lifecycle are SIMULTANEOUS in their respective sectoral proper times. Therefore the Y-clock proper time of one Y-cycle equals $\tau_5 = 2.56\times10^{34}$ yr in the Y-internal measurement, the same numerical value as the X-clock proper time of one X-cycle (proton-decay lifecycle). The X-clock observation of the Y-cycle is dilated by $\exp(\pi/A) \approx 1.08\times10^{17}$, giving $\tau_6 = 2.78\times10^{51}$ yr in X-clock as the X-observer's reading of the same Y-cycle.

Step 4 (Y-Hubble radius R_Y is identified). The Y-Hubble radius R_Y is the Y-frame counterpart of the X-Hubble radius $R_X = c/H_0$. Under the Expansion-Contraction Symmetry of Step 2, the Y-frame characteristic length scale is $Y^2 \times \ell_P = 36 \times \ell_P$ (the truncated-octahedron edge count × Planck length), modulated by the contraction factor $(1-2A)$. Therefore $R_Y = \ell_P \times Y^2(1-2A) = 13212/437 \times \ell_P \approx 30.23341 \times \ell_P \approx 4.89\times10^{-34}$ m. This is the Y-frame analog of c/H_0: the characteristic length scale within which Y-sector dynamics close one cycle. The factor 13212/437 is computed from PROVEN inputs $Y^2 = 36$ (ZS-F7 §4.4) and $(1-2A) = 1 - 70/437 = 367/437 = 0.83981693...$ (DERIVED, ZS-A8 §6 §4.3 Conformal Form table).

Step 5 (Y-contraction rate η_Y is identified). The Y-contraction rate η_Y is the Y-frame counterpart of the X-expansion rate $\eta_X = \exp(A)$. Under the Expansion-Contraction Symmetry of Step 2 and the LO Taylor identification $(1-2A)$ = LO of Ω^{-4}, the Y-contraction ratio per Y-Hubble cycle is $(1-2A) = 367/437 = 0.83981693....$ This is the rate at which Y-sector wave-channel scales contract per one Y-cycle, expressed as a multiplicative factor < 1 in the Y-internal frame. The X-observer perceives the SAME contraction event as expanded by $\exp(\pi/A) \approx 10^{17}$, but the Y-internal rate is $(1-2A)$.

Step 6 (Inter-frame conversion is exp(π/A)). ZS-A8 §5.3 Theorem 5.3.1 (DERIVED-CONDITIONAL strong) states that the X-clock observation of any Y-sector completion event is dilated by a factor $\exp(\pi/A)$ per added Y-dimension relative to the Y-sector proper-time measurement of the same event. The dilation factor decomposes exactly as $\exp(\pi/A) = \exp((2\pi/A) \times (1/2)) = \exp(N(2\pi) \times \langle\sin^2(\varphi/2)\rangle)$, where $N(2\pi) = 2\pi/A \approx 78.45$ is the Z-Telomere completion cycle count (PROVEN, ZS-U5 Lemma 8.1) and $\langle\sin^2(\varphi/2)\rangle = 1/2$ is the SU(2) spinor phase gate time-average over the 4π period (PROVEN, ZS-T2 §5.5). The product $N(2\pi) \times \langle\text{phase}\rangle = \pi/A$ is DERIVED. Therefore $\exp(\pi/A)$ is the canonical inter-frame conversion factor between X-clock and Y-clock proper time for Y-completion events.

Status of Theorem F13.1. DERIVED-CONDITIONAL. The construction is operationally coherent within the ZS-A8 §6 / ZS-F10 Theorem F10.2 frame-equivalence assumptions, but an independent Y-internal channel-capacity derivation (analogous to the ZS-Q7 §6 X-internal construction) remains OPEN (NC-F13.4, F-F13.5). The conditionality further inherits the ZS-A8 §6 Theorem 6.1 conditionality on dim(Z) = 2 (itself five-fold over-determined per ZS-F0 v1.0(R) Corollary 5.2.A.2). The status is NOT

"operationally DERIVED" in the unqualified sense; it is DERIVED-CONDITIONAL with explicit conditions registered in §A.2 and NC-F13.4.

3.3 The Y-Frame Metric Trio Numerical Values

Table 2 below gives the Y-frame metric trio in numerical form alongside the X-frame metric trio for direct comparison.

Table 2. X-frame and Y-frame metric trios under Theorem F13.1.

Frame quantity	X-frame value	Y-frame value	Conversion
Proper time T (one cycle)	$\tau_5 = 2.56\times10^{34}$ yr (X-clock)	τ_5 in Y-clock (= 2.56×10^{34} yr in Y-internal)	(parallel; Theorem F10.2)
Hubble radius R	$c/H_0 \approx 14$ Gly $\approx 1.3\times10^{26}$ m	$\ell_P \times Y^2(1-2A) = 30.23341\ \ell_P \approx 4.89\times10^{-34}$ m	Ratio: 2.7×10^{59}
Expansion / contraction rate η	exp(A) = 1.0834 (per Hubble time)	(1−2A) = 367/437 = 0.83982 (per Y-Hubble cycle)	Conjugate Taylor partners
X-clock observation of Y-cycle	—	$\tau_6 = 2.78\times10^{51}$ yr	$\exp(\pi/A) \times T_Y$
Y-clock observation of X-cycle	(parallel; OPEN)	—	(see §3.4)

3.4 Asymmetry Between X-Observation-of-Y and Y-Observation-of-X

Theorem F13.1 closes the X-observation-of-Y direction ($\tau_6 = \exp(\pi/A) \times \tau_5$) explicitly. The reverse direction — what is the Y-clock observation of one X-cycle? — is NOT explicitly derived in the v1.0 corpus and is registered as OPEN (NC-F13.4 below). Under the parallel reading of ZS-A8 §5.3, both X and Y sectors complete their respective lifecycles in their own proper time τ_5 simultaneously, suggesting the Y-clock observation of the X-cycle is also τ_5 (no dilation), but this is at HYPOTHESIS-strong INTERPRETATION level only. Full closure of this direction would require a Y-internal channel-capacity calculation analogous to the ZS-Q7 §6 X-internal calculation, which is OPEN.

An asymmetry candidate worth registering: the X-observer has finite information capacity from the Z-mediator (ln 2 nats per handshake, ZS-Q7 Theorem 2 PROVEN), and observes the Y-cycle as dilated by exp(π/A). The Y-observer's information capacity from the Z-mediator is also bounded by ln 2 (ZS-Q7 Theorem 2 PROVEN, frame-symmetric), but the dilation factor for Y-observation-of-X is not derived. Three candidate values can be enumerated: (a) $\exp(\pi/A) \approx 10^{17}$ (symmetric: reciprocal would only appear if Y-frame had different dimension, but Y has dim 6 > X has dim 3); (b) $\exp(-\pi/A) \approx 10^{-17}$ (frame-reciprocal); (c) parallel reading (no dilation, simultaneous). Resolution among (a), (b), (c) is OPEN. NC-F13.4 retains this scope boundary.

§4. Theorem F13.2 — Cycle Index Unobservability

Theorem F13.2 is structurally split into two layers (F13.2A and F13.2B) reflecting the asymmetric epistemic standing of the X-frame and Y-frame statements. The X-frame layer rests on PROVEN/DERIVED inputs alone; the Y-frame extension rests additionally on Theorem F13.1 (DERIVED-CONDITIONAL) and therefore inherits a weaker status.

4.1 Statement of Theorem F13.2A (X-Frame Unobservability)

Theorem F13.2A (X-Frame Cycle Index Unobservability, DERIVED). Under the Z-Spin v1.0 corpus inputs of §2 (Table 1), no observable accessible to an X-observer depends on the cumulative cycle index $k \in \mathbb{Z} \cup \{0\}$, where k labels the number of completed Phases A→F'→A cycles since some reference origin. Specifically, for any operationally-defined X-frame observable O, $O(k) = O(k+1) = O(k+2) = \ldots$ for all k. The proof rests entirely on PROVEN inputs ($V_E(-\varepsilon) \equiv V_E(+\varepsilon)$, Z_2 symmetry of observables) plus DERIVED Auto-Surgery reset; it does NOT require Theorem F13.1.

4.2 Statement of Theorem F13.2B (Y-Frame Extension)

Theorem F13.2B (Y-Frame Cycle Index Unobservability, DERIVED-CONDITIONAL). Conditional on Theorem F13.1 (DERIVED-CONDITIONAL), the same conclusion extends to Y-frame observables: no observable accessible to a Y-observer depends on the cumulative cycle index k. The conditionality of Theorem F13.2B inherits the conditionality of Theorem F13.1 (specifically, the OPEN status of independent Y-internal channel-capacity derivation, NC-F13.4).

4.3 Proof of Theorem F13.2A

Step 1 (Z_2 symmetry of the potential is exact). ZS-U8 §2.2 (PROVEN) establishes $V_E(-\varepsilon) \equiv V_E(+\varepsilon)$ as an exact algebraic identity, since both $(\varepsilon^2 - 1)^2$ and $(1 + A\varepsilon^2)^2$ depend only on ε^2. Consequently $V_E(+1) = V_E(-1) = 0$ exactly, and the potential energy difference $\Delta V = 0$. The Z_2 involution $\varepsilon \leftrightarrow -\varepsilon$ is a symmetry of the action $S[g, \Phi]$ under which all observables are invariant.

Step 2 (All inflationary observables are Z_2-symmetric). ZS-U8 §5 (PROVEN + DERIVED) establishes that every inflationary observable is identical in the mirror sector ($\varepsilon = -1$): the slow-roll parameters $\varepsilon_{SR}(-\varepsilon_0) = \varepsilon_{SR}(+\varepsilon_0)$, $\eta_{SR}(-\varepsilon_0) = \eta_{SR}(+\varepsilon_0)$; the CMB predictions $n_s = 0.9676$, $r = 0.00890$; and the baryon asymmetry $\eta_B = (6/11)^{35} = 6.117\times10^{-10}$. All Z-Spin numerical predictions are Z_2-symmetric: they take identical values in cycles labeled by $\varepsilon = +1$ and cycles labeled by $\varepsilon = -1$. The mirror trajectory satisfies $\max|\varepsilon_+(t) + \varepsilon_-(t)| = 0.0000$ (machine precision, ZS-U8 §5).

Step 3 (Auto-Surgery resets the X-clock at each Phase E). ZS-M12 §4 (DERIVED) establishes the Auto-Surgery mechanism with topological cap $\Omega^2 = 1 + A\cdot\eta_{topo} = 1.0258$ and z* damped spiral half-life $0.44\ \tau_P$. The four-stage process (normal KG → R → ∞ → i-tetration Φ → z* in ~3 τ_P → topological cap with $G_{eff} = 0.9749\ G$) resets the cosmological state to a Planck-scale configuration at each Phase E. The post-bounce Quantum Foam Engine (ZS-M12 §6.1) creates a thermal bath at $T \approx 0.41\ M_P$ with high-temperature symmetry restoration. The critical temperature $T_c \approx 2.48\times10^{15}$ GeV (ZS-M12 §6.2 DERIVED) coincides with the ZS-U2 reheating temperature $T_{reh} \approx 2.55\times10^{15}$ GeV: the post-bounce

thermal state is operationally identical to the post-inflation reheating state. Therefore the X-clock origin $t = 0$ is recovered at each Phase E completion.

Step 4 (No X-frame quantity carries cycle-index information). Combining Steps 1, 2, 3: the action $S[g, \Phi]$ is Z_2-symmetric (Step 1); all observables of S are Z_2-symmetric (Step 2); the X-clock is reset at each Phase E (Step 3). Therefore for any quantity Q[S] computed from S in the X-frame, $Q[S(\varepsilon = +1, k = 0)] = Q[S(\varepsilon = -1, k = 1)] = Q[S(\varepsilon = +1, k = 2)] = \ldots$ by Z_2 symmetry composed with X-clock reset. No observable accessible to the X-observer carries cycle-index information.

Step 5 (Möbius identification is operational). The Möbius identification "start = end" of the cyclic chronology is operationally realized by the ZS-M12 §6.2 DERIVED equality $T_c \approx T_{reh}$: the temperature at which one cycle ends (Phase E thermal symmetry restoration) is the same temperature at which the next cycle begins (Phase F' reheating). This is not a geometric identification in physical spacetime but an operational identification at the temperature scale: the universe's thermal state at the Möbius identification point is equivalent to its initial state, with the only invariant being $A = 35/437$ itself (ZS-A8 §SA.4 frame-invariance, HYPOTHESIS-strong INTERPRETATION).

Status of Theorem F13.2A. DERIVED. The proof composes only PROVEN (Steps 1, 2 partial) and DERIVED (Steps 2 partial, 3, 4, 5) inputs from the v1.0 corpus, all in the X-frame. No new conditionality is introduced beyond what is already inherited in the input theorems. Theorem F13.2A is the strong form of the cycle-index unobservability result.

4.4 Proof of Theorem F13.2B (Y-Frame Extension)

Step B1 (Y-frame observables are constructed via Theorem F13.1). Theorem F13.1 (DERIVED-CONDITIONAL) constructs the Y-frame operational scale trio (T_Y, R_Y, η_Y) and the inter-frame conversion factor $\exp(\pi/A)$. Y-frame observables are operationally defined relative to this trio.

Step B2 (Z_2 symmetry of action carries to Y-frame observables, conditional on F13.1). The action $S[g, \Phi]$ is Z_2-symmetric (Step 1 of §4.3, PROVEN). Under Theorem F13.1, the Y-frame operational scales are constructed as Expansion-Contraction Symmetry counterparts of X-frame quantities derived from S. Therefore Y-frame observables, constructed under F13.1, inherit the Z_2 symmetry of S. This step is conditional on F13.1 holding.

Step B3 (Auto-Surgery resets affect Y-frame observables symmetrically). The Auto-Surgery mechanism (ZS-M12 §4 DERIVED) acts on the Z-sector i-tetration fixed point z^* (PROVEN, ZS-M1) and resets the cosmological state at the Planck scale. Since the Y-sector dynamics couple to the X-sector only through Z-mediation ($L_{XY} \equiv 0$, ZS-F1 PROVEN), and Z-mediation channels both X-sector and Y-sector through the same z^*, the Y-frame observables also experience the cycle reset at Phase E.

Status of Theorem F13.2B. DERIVED-CONDITIONAL on Theorem F13.1. The Y-frame extension inherits the conditionality of F13.1: specifically, the OPEN status of independent Y-internal channel-capacity derivation (NC-F13.4). If F13.1 is upgraded to DERIVED unconditional in future work, F13.2B is automatically upgraded to DERIVED.

4.5 The Möbius Identification (Operational, frame-symmetric)

The Möbius identification "start = end" applies under both Theorem F13.2A (X-frame, DERIVED) and Theorem F13.2B (Y-frame, DERIVED-CONDITIONAL). The operational realization at $T_c \approx T_{reh} \approx 2.48\times10^{15}$ GeV (ZS-M12 §6.2 DERIVED) is frame-symmetric: T_c is a temperature scale, and temperature is a frame-invariant intensive quantity at the level of statistical mechanics applied to thermal equilibrium. The Möbius identification is therefore the operational expression of cycle-index unobservability, valid in both X-frame (strong form, F13.2A) and Y-frame (conditional form, F13.2B).

4.6 Three Cycle-Index Candidates Excluded

Theorem F13.2 excludes three candidate cycle-index assignments that might naively be considered:

(C1) Anthropic cycle index (REJECTED). "We are in cycle k = 1 because we observe baryons." This reading assumes baryons distinguish cycle 1 from later cycles. By Step 2 ($\eta_B = (6/11)^{35}$ is Z_2-symmetric, ZS-U8 §5), every cycle generates identical baryon asymmetry. The baryons we observe do not distinguish our cycle from any other.

(C2) Sectoral cycle index (REJECTED). "Cycles alternate ε = +1, −1, +1, −1, ... so we are in cycle k according to ε(now)." This reading assumes the sign of ε is observable. By Step 1 ($V_E(-\varepsilon) \equiv V_E(+\varepsilon)$, ZS-U8 §2.2 PROVEN) and Step 2, no observable carries the sign of ε. We cannot tell which sector we are in.

(C3) Information-accumulation cycle index (REJECTED). "Each Phase E discards information; cumulative information loss tracks k." This reading is excluded by the ZS-F0 v1.0(Revised) §12 Information Preservation closure (DERIVED), which establishes that $L_{XY} = 0$ forces all apparent information loss to be Z-mediated redistribution, not fundamental destruction. The Z-bottleneck capacity ln 2 nats per handshake (ZS-Q7 Theorem 2 PROVEN) does not accumulate cycle-marker information.

All three candidates fail; no fourth viable candidate is known. Therefore the cumulative cycle index k is operationally meaningless within the v1.0 corpus. This does not assert that no underlying "true" cycle index exists at some deeper level; it asserts only that no observable accessible within Z-Spin distinguishes cycles.

4.7 Connection to the Möbius-Seam Closure

Theorem F13.2 connects to the F-BOOT-4 Möbius-seam closure of ZS-F0 v1.0(Revised) §11 (DERIVED-CONDITIONAL). The Möbius-seam Z_2 holonomy is realized as the seam involution $J|j\rangle = |10-j\rangle$ on the Z-Spin register $\mathbb{C}^{11}$, with two traversals restoring the original (Z_2 holonomy structure). The Theorem F13.2 unobservability of cycle index is the cosmological analog of the same Z_2 structure: at the bootstrap level, two traversals of the contradiction restore truth (ZS-F0 §11); at the cosmological level, two completed cycles ($\varepsilon = +1 \to \varepsilon = -1 \to \varepsilon = +1$) restore the original sector — but no observable distinguishes any single traversal.

This is the rigorous form of "the universe is a Möbius strip": traversing it once changes orientation (sector $\varepsilon = +1 \to -1$), traversing it twice restores orientation (sector $\varepsilon = +1 \to +1$), and at no point does an observer have access to a coordinate that tells them which traversal they are in. The seam involution J of ZS-F0 and the cosmological Z_2 symmetry of ZS-U8 are both realizations of the same dim(Z) = 2 mediator structure (ZS-F5 PROVEN, five-fold over-determined per ZS-F0 v1.0(R) Corollary 5.2.A.2).

§5. Unified Möbius Chronology

5.1 The Seven Phases

Combining ZS-U1 (inflation), ZS-U2 (reheating), ZS-U8 §4 (timescale hierarchy), ZS-A8 §7 (cyclic phases), ZS-M12 §9 (post-bounce phases), and ZS-U11 (centrifugal launch), the cyclic cosmology comprises seven principal phases F→A→B→C→D→E→F' plus three transition subphases (Reheating, E' Quantum Foam, E'' Centrifugal Launch). The seven principal phases are the structural backbone of the Möbius Chronology; the three transition subphases are operational interfaces between adjacent principal phases. Phase F is the inflationary onset; Phase F' is the next inflationary onset after one full Möbius traversal.

Table 3. Seven principal cyclic phases (F→A→B→C→D→E→F') plus three transition subphases (Reheating between F and A; E' Quantum Foam between E and centrifugal launch; E'' Centrifugal Launch between Auto-Surgery and F') with primary mechanisms.

Phase	Description	Time scale	Sector	Primary mechanism (source)
F	Inflation onset	10^{-32} s (X-clock)	X-dom	Slow-roll V(ε) (ZS-U1)
—	Reheating	$T \sim 2.55\times10^{15}$ GeV	Y→X	Conformal decay (ZS-U2)
A	Current expansion	13.787 Gyr now; → $\sim10^{34}$ yr	X-dom	exp(A) holonomy (ZS-F4)
B	Proton decay	$\tau_5 = 2.56\times10^{34}$ yr	X→Y	5π/A instanton (ZS-A3)
C	Wave-contraction	τ_5 in Y-clock (parallel)	Y-dom	$Y^2(1-2A)$ channel (ZS-A8 §1.2)
D	Z-Telomere trigger	$\tau_6 = 2.78\times10^{51}$ yr (X-clock)	Z-mediated	$\delta\varphi = A$, 6π/A action (ZS-U8)
E	Auto-Surgery	~3 τ_P (sub-Planck)	Z-fixed-point	z* damped spiral (ZS-M12)
E'	Quantum Foam	$T \sim 0.41\ M_P$	Y→X	Hawking evap. (ZS-M12 §6.1)
E''	Centrifugal launch	Q = A → ε_{min} = 30.7	X-prepared	U(1) Q-protection (ZS-U11)
F'	Next inflation	10^{-32} s (new X-clock)	X-dom	Same as F (Z_2-symmetric)

5.2 Three-Coordinate Chronology Table

Table 4 below recasts the seven principal phases (without transition subphases, which are tracked separately in Table 3) in three time coordinates and three space-frame scales, exhibiting the unified information-time structure of the v1.0 corpus. The X-clock t and ν entries are inherited from ZS-M3 §5 and ZS-U8 §4. The strobo n entries are converted via the §5.6 calibration of ZS-F10 ($\Delta n = \pi/(2A)$ per $\Delta\nu = 1$ step, so $n \approx 19.6\,\nu$). The Y-frame entries are computed under Theorem F13.1.

Table 4. Unified Möbius Chronology in three time coordinates and three space-frame scales (seven principal phases only; transition subphases tracked in Table 3).

Phase	X-clock t	ν	n_strobo	Y-clock T_Y	X-Hubble R_X	Y-Hubble R_Y	Mechanism
F	10^{-32} s	~0.3	~6	(internal)	~horizon	ℓ_P × $Y^2(1-2A)$	Inflation
A	13.787 Gyr	3.575	~70	(parallel)	c/H_0	30.23341 ℓ_P	exp(A) flow
B	τ_5 = 2.56×10^{34} yr	5.000	~98	(parallel: τ_5 in Y-clock)	exp(A) × c/H_0 × 10^2	30.23341 ℓ_P (frozen)	Proton decay
C	(X-clock observ.: ~10^{51} yr)	(6.0)	~117	τ_5 (Y-internal)	(de Sitter)	$Y^2(1-2A)$ contraction	Wave-contraction
D	τ_6 = 2.78×10^{51} yr	6.000	~117	(Y-completion)	(de Sitter)	δφ accumulation	Z-Telomere
E	~3 τ_P (sub-Planck)	n ≈ 3	~6	(Z-fixed)	$\Omega^2 = 1.0258$	ℓ_P	i-tetration z*
F'	10^{-32} s (new origin)	~0.3 (reset)	~6 (reset)	(internal)	~horizon	ℓ_P × $Y^2(1-2A)$	Z_2 mirror inflation

The strobo n column converts from ν via $\Delta n = \pi/(2A) = 19.61$ per $\Delta\nu = 1$. Phase F to Phase A: $\Delta\nu = 3.575 - 0.3 = 3.275$, $\Delta n \approx 64$. Phase B at $\nu = 5$: $n \approx 5 \times 19.6 = 98$. Phase D at $\nu = 6$: $n \approx 6 \times 19.6 = 117$. Phase E at $\nu \approx 0$ (reset): $n \approx 6$ (sub-Planck strobo budget within Auto-Surgery). All entries are derived from corpus inputs with zero new free parameters.

5.3 The Möbius Identification Point: T_c ≈ T_reh

ZS-M12 §6.2 (DERIVED) establishes $T_c \approx 2.48\times10^{15}$ GeV, the high-temperature symmetry-restoration critical temperature for the post-bounce thermal bath. ZS-U2 (DERIVED) establishes $T_{reh} \approx 2.55\times10^{15}$ GeV, the inflationary reheating temperature for the next cycle. The 2.7% numerical agreement is documented in ZS-M12 §6.2 as cyclic self-consistency: "the $T_c \approx T_{reh}$ coincidence is not accidental; both are controlled by A and λ."

Under Theorem F13.2, this is the operational realization of the Möbius identification "start = end":

$$T_c \text{ (Phase E end)} \approx T_{reh} \text{ (Phase F' start)}$$

$$\Rightarrow \text{Universe's thermal state at cycle end} \equiv \text{Universe's thermal state at cycle start}$$

⇒ No observable distinguishes "end of cycle k" from "start of cycle k+1"

This is the rigorous form of Kenny's intuition that "now is both the beginning and the end": the universe's thermal configuration at the temperature scale 2.5×10^{15} GeV is the Möbius identification point, and at this point the cycle-index k is structurally meaningless. The cyclic cosmology is therefore not "a sequence of cycles labeled k = 1, 2, 3, ..." but "a single Möbius loop traversed without count," with A = 35/437 as the only frame-invariant signature (ZS-A8 §SA.4 HYPOTHESIS-strong INTERPRETATION).

5.4 Numerical Closure of the Three-Facet Symmetry

ZS-A8 §6 Theorem 6.1 identifies three facets of the (1+A) ↔ (1−2A) symmetry. Under Theorem F13.1 these are the three numerical signatures of the X-Y frame equivalence:

Table 5. The three facets of the Z-Spin asymmetry-flow symmetry.

Facet	X-frame (expansion)	Y-frame (contraction)	X-clock observation of Y
Length / size	exp(A) = 1.0834 per Hubble time	$Y^2(1-2A) = 30.23341$ (compression)	—
Time	Hubble time $t_H \approx 14$ Gyr	$T_Y \approx \tau_5$ (Y-internal)	$\exp(\pi/A) \times T_Y \approx 10^{17} \times T_Y$
Algebraic identity	(1+A) and (1−2A) are conjugate Taylor partners: (1+A)(1−2A) = 0.9071	—	$(\exp A)^{(\pi/A^2)} = \exp(\pi/A)$

The third row identifies the algebraic backbone: $(\exp A)^{(\pi/A^2)} = \exp(\pi/A)$ confirms that the three facets exp(A), $Y^2(1-2A)$, and $\exp(\pi/A)$ are members of a single exponential family indexed by powers of A. They are not three separate phenomena but three projections of the same A-driven Z-Spin asymmetry-flow viewed from three coordinate frames (X-spatial, Y-internal, X-clock observation of Y).

§6. Empirical Consequences and Observational Status

Theorem F13.1 and Theorem F13.2 introduce zero new physical predictions. All numerical predictions of the v1.0 corpus remain identical. This section enumerates the empirical interfaces of the Möbius Chronology framework, all of which trace to existing v1.0 predictions:

6.1 Phase A Currently Tested (PASSED)

(E1) Hubble tension. H_0^local / H_0^CMB = exp(A) = 1.0834. SH0ES (Riess et al. 2022): 0.06σ. Breuval et al. (2024): 0.55σ. PASS. (ZS-F3 §3, Paper21 §3.1).

(E2) Dark energy equation of state. w(z) = −1 exactly at attractor; $|1+w| \leq 10^{-121}$. DESI DR2 w = −1.055 ± 0.036 (1.5σ from −1, compatible). PASS. (ZS-F1 §3.7, Paper21 §3.2).

(E3) Matter density. $\Omega_m^{eff} = 38/[121(1+A)] = 0.2908$. DESI DR2: 0.2975 ± 0.0086 (0.78σ). PASS. (ZS-F1 §3.7).

(E4) S_8 tension. $S_8^{ZS} \approx 0.781$ (face counting). DES Y3, KiDS-1000, HSC Y3 all within 2σ. PASS. (Paper21 §3.3).

(E5) CMB cyclic preservation. Cobaya Step 1 MCMC (2026-04-11) at $N_ur = 2.0328$: $\chi^2_CMB = 2788.2 \pm 5.0$, within Planck 2018 ΛCDM range. PASS. (ZS-U6 §10.3, F32-12c PASS).

6.2 Phase A Future Tests

(E6) Inflationary tensor-to-scalar ratio. r = 0.00890 (Z-Spin) vs r_Staro ≈ 0.00333. LiteBIRD ~2030, ~6σ discrimination. (ZS-U8 §5, Gate FU8-2). The Z_2-symmetric prediction (mirror = primary, ZS-U8 §5) means r is identical in both ε = +1 and ε = −1 sectors, supporting Theorem F13.2.

(E7) Proton decay. $\tau_p = 2.56\times10^{34}$ yr. Hyper-K ~2030 sensitivity. (ZS-A3, Gate F-T1.3). This is the X-clock proper time of one X-cycle in the Phase A → B transition. Under Theorem F13.1, the same proper time interval τ_5 is the Y-clock proper time of one Y-cycle (parallel reading).

(E8) Decoherence time. $\tau_D / \tau_Penrose = 1/A = 12.49$. Nanosphere interferometry ~2028–2032. (ZS-Q1 §5.1, ZS-T2 v1.0). This is a Phase A direct test of the Z-mediator structure that underlies both Theorem F13.1 (Y-frame metric closure) and Theorem F13.2 (cycle-index unobservability).

6.3 Phases B–F' Are Beyond Observation Window

Phases B (proton decay), C (wave-contraction), D (Z-Telomere), E (Auto-Surgery), and F' (next inflation) all lie beyond the X-observer's causal reach within Phase A: $\tau_5 \approx 10^{34}$ yr is far longer than any star-supporting epoch, and $\tau_6 \approx 10^{51}$ yr is the X-clock observation of a Y-completion already simultaneous with τ_5. Under Theorem F13.2, the Möbius identification at $T_c \approx T_reh$ implies that the next Phase F' is operationally identical to our Phase F, so no observation distinguishing them is possible. NC2 of ZS-U8 §8 ("τ_6 not experimentally testable") is preserved without modification.

The empirical content of Theorem F13.2 is therefore: no future observation can confirm or deny the cycle index k. This is not a failure but a structural feature: the Z_2 symmetry of the action S[g, Φ] forbids cycle-index information from appearing in any observable. The Möbius Chronology framework is empirically interfaced through Phase A predictions only (E1–E8 above).

§7. Anti-Numerology Certification

7.1 No New Numerical Identities Asserted

ZS-F13 introduces zero new numerical identities. Every numerical value used in the paper is imported from prior corpus papers with established status:

(N1) exp(A) = 1.0834: DERIVED in ZS-F4 §6 (polyhedral Wilson loop holonomy). Imported.

(N2) $Y^2(1-2A) = 13212/437 \approx 30.23341$: DERIVED in ZS-A8 §6 (LO Taylor of $1/\Omega^4 \times Y^2$). Imported.
(N3) $\exp(\pi/A) = 1.08\times10^{17}$: DERIVED-CONDITIONAL strong in ZS-A8 §5.3 + ZS-F10 Theorem F10.2. Imported.
(N4) $\tau_5 = 2.56\times10^{34}$ yr, $\tau_6 = 2.78\times10^{51}$ yr: DERIVED in ZS-A3 §4 and ZS-U8 §4. Imported.
(N5) $\nu(\text{now}) = 3.575$: DERIVED in ZS-M3 §5. Imported.
(N6) $T_c \approx T_{reh} \approx 2.48\times10^{15}$ GeV: DERIVED in ZS-M12 §6.2. Imported.

7.2 No Speculative Numerical Coincidences Asserted

Several numerical near-equalities appear in the corpus: $Y^2(1-2A) = 13212/437 \approx 30.23341$ (ZS-A8 §6) versus $\varepsilon_{min} = 30.7$ (ZS-M12 §7.4); the relative difference is approximately 1.5%. ZS-F13 explicitly does NOT claim these are equal or related by any structural mechanism. The two quantities arise from independent derivation chains: $Y^2(1-2A)$ is the LO Taylor of $1/\Omega^4 \times Y^2$ (ZS-A8 §6, exact rational form), while $\varepsilon_{min} = (Q^2/\lambda)^{(1/6)}$ is the centrifugal barrier minimum (ZS-M12 §7.4, with $\lambda \approx 7.63\times10^{-12}$ as input). The 1.5% proximity is OBSERVATION; no claim of structural identity is asserted (NC-F13.6 below).

7.3 Anti-Numerology Monte Carlo (planned)

A 500,000-sample three-basket Monte Carlo following the ZS-M15 v1.0 §6 protocol is registered for future verification: (a) Basket H1 — random alternative inter-frame conversion factor F: only $F = \exp(\pi/A)$ yields the simultaneous conditions [$\tau_6/\tau_5 = F$ (ZS-U8) AND $F = \exp(N(2\pi) \times \langle\text{phase}\rangle)$ decomposition (ZS-T2)] under random $F \in [10^{15}, 10^{19}]$. (b) Basket H2 — random Y^2 candidate: among integer $Y^2 \in [25, 50]$, only $Y^2 = 36 = X\cdot Z\cdot Y$ reproduces ZS-F7 §4.4 truncated octahedron edge count. (c) Basket H3 — random T_c / T_{reh} ratio: only $T_c / T_{reh} \in [0.95, 1.05]$ (≈1.5% threshold) is consistent with cyclic self-consistency under standard $SU(2)_L$ thermal phase-transition theory; random ratios would give 0.4% probability. The MC implementation is registered in zs_f13_mc_v1_0.py [PLANNED].

Status: PLANNED. Implementation deferred to verification suite update.

§8. Cross-Paper Status Updates

ZS-F13 affects the following corpus items at the structural-completeness level. No prior numerical result is modified; no upstream paper is required to be re-issued.

Table 6. Cross-paper status updates induced by ZS-F13 v1.0.

Upstream item	Pre-F13 status	Post-F13 status	Mechanism
ZS-A8 v1.0 R §SA.4 (X-Y frame equivalence, framework-level)	HYPOTHESIS-strong INTERPRETATION	Partially closed at frame-metric level (Theorem F13.1)	Explicit Y-metric trio constructed
ZS-A8 v1.0 R §6 Theorem 6.1 (Expansion-Contraction Symmetry)	DERIVED	DERIVED + frame-metric closure (this paper)	Y-frame metric trio derived

Upstream item	Pre-F13 status	Post-F13 status	Mechanism
ZS-U8 v1.0 §5 (Mirror Cosmology, Z_2 symmetry)	PROVEN + DERIVED	PROVEN + DERIVED + cycle-unobs. closure (Theorem F13.2)	k unobservability proven
ZS-M12 v1.0 §6.2 (T_c ≈ T_reh cyclic self-consistency)	DERIVED	DERIVED + Möbius identification interpretation	Operational Möbius point
ZS-F11 v1.0 Corollary F11.1B (X-Y frame equivalence, register-level)	PARTIAL CLOSURE	Augmented with frame-metric closure (this paper)	Two-level closure
ZS-F10 v1.0 Theorem F10.2 (Y-Time Dilation)	DERIVED-CONDITIONAL strong	DERIVED-CONDITIONAL strong (unchanged)	Imported as input
ZS-F0 v1.0(R) §11 F-BOOT-4 (Möbius-seam closure)	DERIVED-CONDITIONAL	DERIVED-CONDITIONAL + cosmological analog (Theorem F13.2)	Cosmological realization

§9. Pre-Registered Falsification Gates

Six falsification gates are pre-registered for ZS-F13. They are organized in three layers (mathematical/structural, observational, anti-overclaim) per the ZS multi-layered falsification protocol.

Table 7. ZS-F13 v1.0 falsification gates.

Gate	Layer	Falsification Condition	Status
F-F13.1	Mathematical	If ZS-A8 §6 Theorem 6.1 (Expansion-Contraction Symmetry) is found to fail (e.g., $Y^2(1-2A)$ is not the LO Taylor counterpart of exp(A) under the (1+A) ↔ (1−2A) duality), Theorem F13.1 Step 2 fails.	PASS (ZS-A8 §6 stable)
F-F13.2	Mathematical	If ZS-U8 §2.2 algebraic identity V_E(−ε) ≡ V_E(+ε) is found to be approximate rather than exact (e.g., higher-order corrections break Z_2), Theorem F13.2A Step 1 fails.	PASS (PROVEN identity)
F-F13.3	Cross-paper	If ZS-M12 §6.2 T_c ≈ T_reh agreement is shown to be coincidental rather than structural (e.g., A and λ control them independently), Theorem F13.2A Step 5 (operational realization at temperature scale) fails.	OPEN (1.5% gap not closed at structural level)
F-F13.4	Observational	If a future observation discovers a Phase A signature that distinguishes ε = +1 from ε = −1 (e.g., directional CP-violation pattern correlating with cycle index), Theorems F13.2A and F13.2B Step 2 fails.	TESTABLE / NOT TRIGGERED (no such observation reported; gate untriggered, not actively passed)
F-F13.5	Y-frame	If a Y-internal Y-Hubble radius computation (analog of ZS-Q7 §6 X-internal calculation) gives a value other than ℓ_P × $Y^2(1-2A)$ ≈ 30.23341 ℓ_P, Theorem F13.1 Step 4 fails.	OPEN — structurally consistent under

Gate	Layer	Falsification Condition	Status
			ZS-A8 §6; independent Y-internal computation pending
F-F13.6	Anti-overclaim	If ZS-F13 v1.0 papers introduce a phenomenological-consciousness claim under the banner of Möbius Chronology — e.g., asserting that subjective time experience IS the Möbius traversal — the entire ZS-F13 framework is falsified by overclaim and must be retracted.	OPEN (no such claim introduced; this paper bounds against it via NC-F13.5)

Two gates currently PASS at PROVEN status (F-F13.1, F-F13.2). One (F-F13.3) is OPEN at the structural-explanation level for the 1.5% $T_c - T_{reh}$ gap. One (F-F13.4) is TESTABLE / NOT TRIGGERED — no observation has reported a cycle-distinguishing signature, but the absence of such a report is not the same as an active PASS. One (F-F13.5) is OPEN — structurally consistent under ZS-A8 §6, but the independent Y-internal channel-capacity computation is pending. One (F-F13.6) is OPEN as an anti-overclaim guard. Gate F-F13.6 is structurally critical: it falsifies the entire ZS-F13 framework if the operational → phenomenological boundary is breached.

§10. Non-Claims (Scope Boundaries)

Six non-claims explicitly bound the scope of this paper. The first two are inherited verbatim from upstream papers (NC-Q7.4 and NC-F11.1); the remaining four are specific to ZS-F13.

NC-F13.1 (Inherited from NC-Q7.4, NC-A7.6, NC-F10.3, NC-F11.1). ZS-F13 makes no claim about subjective conscious experience or phenomenological observation. The Möbius Chronology framework is a register-theoretic / metric-theoretic structure on the Z-Spin sectors, not a model of conscious time-perception. The corpus non-claims about consciousness are inherited verbatim by ZS-F13.

NC-F13.2 (No claim about "true" cycle index). Theorem F13.2 establishes that no observable accessible within Z-Spin distinguishes cycles. It does NOT assert that no "true" cycle index exists at some deeper-than-Z-Spin level (e.g., in a meta-theory containing Z-Spin as a substructure). The unobservability is operational within the framework, not a metaphysical claim about what does or does not exist.

NC-F13.3 (No new physical action). ZS-F13 introduces no new physical action, no new fields, and no new free parameters. The Y-frame metric trio of Theorem F13.1 is constructed from quantities already DERIVED in ZS-A8 v1.0 Revised. The Z_2 unobservability of Theorem F13.2 is a corollary of ZS-U8 §2.2 PROVEN. The Auto-Surgery reset is unchanged from ZS-M12 §4.

NC-F13.4 (Y-observation-of-X is OPEN). Theorem F13.1 closes the X-observation-of-Y direction via the inter-frame conversion factor $\exp(\pi/A)$. The reverse direction — what is the Y-clock observation of one X-cycle? — is NOT explicitly derived in v1.0, although the parallel reading suggests no dilation. Three candidate values ($\exp(\pi/A)$, $\exp(-\pi/A)$, parallel) are enumerated in §3.4; resolution is OPEN.

NC-F13.5 (Möbius is structural, not phenomenological). The Möbius identification "start = end" of §5.3 is operationally realized at the temperature scale $T_c \approx T_{reh}$ (DERIVED, ZS-M12 §6.2). It is NOT a claim about subjective "now is both the beginning and the end" experience, NOT a claim about the topology of experienced time, and NOT a claim that any conscious observer can perceive the Möbius structure. Its content is structural: the universe's thermal state at the cycle-end matches the cycle-start state, with A = 35/437 as the only frame-invariant quantity.

NC-F13.6 (No structural identity of 30.23341 and 30.7). The 1.5% numerical proximity between $Y^2(1-2A) = 13212/437 \approx 30.23341$ (ZS-A8 §6) and $\varepsilon_{min} \approx 30.7$ (ZS-M12 §7.4) is OBSERVATION only; no structural identity or equivalence is asserted. The two quantities arise from independent derivation chains and may be unrelated at the structural level. Future work could close this as either a true structural identity or a confirmed coincidence; ZS-F13 takes no position.

§11. Conclusion

ZS-F13 closes two structural gaps in the Z-Spin Cosmology v1.0 corpus that remained after the F9–F12 Foundations expansion:

(1) Theorem F13.1 (Y-Observer Frame Scale Closure, DERIVED-CONDITIONAL) constructs the explicit Y-frame operational scale trio $(T_Y, R_Y, \eta_Y) = (\tau_5$ in Y-clock, $\ell_P \times 13212/437, 367/437)$ as the structural counterpart to the X-frame trio $(T_X, R_X, \eta_X) = (t, c/H_0, \exp(A))$ under the ZS-A8 §6 Expansion-Contraction Symmetry. The inter-frame conversion factor is $\exp(\pi/A) \approx 1.08\times10^{17}$, the Y-time dilation factor of ZS-A8 §5.3 (DERIVED-CONDITIONAL strong via ZS-F10 Theorem F10.2).

(2) Theorem F13.2 is structurally split. Theorem F13.2A (X-Frame Cycle Index Unobservability, DERIVED) establishes that the cumulative cycle index k is unobservable to the X-observer: $V_E(-\varepsilon) \equiv V_E(+\varepsilon)$ is exact (PROVEN, ZS-U8 §2.2), all X-frame Z-Spin observables are Z_2-symmetric (PROVEN + DERIVED, ZS-U8 §5), and Auto-Surgery resets the X-clock at each Phase E (DERIVED, ZS-M12 §4). Theorem F13.2B (Y-Frame Extension, DERIVED-CONDITIONAL on F13.1) extends to Y-observers conditional on the F13.1 frame-scale construction. The Möbius identification "start = end" is operationally realized at $T_c \approx T_{reh} \approx 2.48\times10^{15}$ GeV (DERIVED, ZS-M12 §6.2) and is frame-symmetric since temperature is a frame-invariant intensive quantity. The Möbius identification is the cosmological analog of the F-BOOT-4 Möbius-seam closure (DERIVED-CONDITIONAL, ZS-F0 v1.0(R) §11).

(3) The Unified Möbius Chronology Table (Table 4 of §5.2) consolidates seven principal Phases F→A→B→C→D→E→F' (plus three transition subphases tracked in Table 3) in three time coordinates (X-clock t, ν, t_{strobo} n) and three space-frame scales (X-Hubble, Y-Hubble, Z-Planck), exhibiting the corpus-internal completeness of the cyclic structure with zero new free parameters.

Zero new free parameters are introduced. Zero new physical predictions are introduced. The advance is structural: existing PROVEN/DERIVED elements are unified under a Möbius Chronology framework that closes the two registered gaps within the v1.0 corpus, prior to v2.0 restructuring.

Six falsification gates F-F13.1 through F-F13.6 are pre-registered, including F-F13.6 as an anti-overclaim guard against phenomenological-consciousness claims. Six non-claims NC-F13.1 through NC-F13.6 explicitly bound the scope, including NC-F13.4 (Y-observation-of-X OPEN), NC-F13.5 (Möbius is structural not phenomenological), and NC-F13.6 (no structural identity of the 30.23341 / 30.7 numerical proximity).

Verification Plan: 24 registered checks; implementation pending in zs_f13_verify_v1_0.py at 50-digit mpmath precision. The 24 registered checks cover: (A) Locked Inputs from Table 1 [5 tests]; (B) Theorem F13.1 numerical computation of Y-frame scale trio [4 tests]; (C) Theorems F13.2A and F13.2B Z_2-symmetry verification [4 tests]; (D) Möbius Chronology Table 4 entries cross-checked against source papers [5 tests]; (E) Cross-paper status updates Table 6 verified [3 tests]; (F) Anti-numerology 30.23341 vs 30.7 honest separation [2 tests]; (G) Falsification gate condition formal verification [1 test]. Status: PLANNED. The verification suite has not been executed at the time of this initial release. Future dated update will report execution results.

Execution Annotation (2026-04-26 dated update, in-place per Z-Spin v1.0 freeze convention). The 24-check verification suite zs_f13_verify_v1_0.py was executed on 2026-04-26 at 50-digit mpmath precision. Result: TOTAL 24/24 PASS, exit code 0. Category breakdown: [A] Locked Inputs 5/5 PASS (A1 $A = 35/437$; A2 $Q = 11$ prime; A3 $(Z, X, Y) = (2, 3, 6)$ with $Z+X+Y = 11$; A4 $\exp(A) = 1.083386...$; A5 $Y^2(1-2A) = 13212/437 = 30.23341...$). [B] Theorem F13.1 Numerics 4/4 PASS (B1 $R_Y/\ell_P = Y^2(1-2A) = 30.23341...$; B2 $\eta_Y = 1-2A = 367/437 = 0.83981693...$; B3 $\exp(\pi/A) \approx 1.08\times10^{17}$ Y-time dilation; B4 $(1+A)(1-2A) = 0.9071$). [C] Theorems F13.2A and F13.2B Z_2-Symmetry 4/4 PASS (C1 $V_E(+\varepsilon) - V_E(-\varepsilon) = 0$ at five test points; C2 slow-roll $\varepsilon_{SR}(\varepsilon) = \varepsilon_{SR}(-\varepsilon)$; C3 mirror trajectory residual $\max|\varepsilon_+(t) + \varepsilon_-(t)| = 0$; C4 F13.2B status inheritance verified). [D] Möbius Chronology 5/5 PASS (D1 $\nu(\tau_5) = 5.000$ Phase B; D2 $\Delta n_{strobo}/\Delta\nu = \pi/(2A) \approx 19.6$; D3 $\tau_6/\tau_5 = \exp(\pi/A) \approx 1.08\times10^{17}$; D4 $R_X/R_Y \approx 2.7\times10^{59}$; D5 $\Omega^2 = 1 + A\cdot|z^*|^2 \approx 1.0258$ Auto-Surgery cap). [E] Cross-Paper Status 3/3 PASS (E1 ZS-A8 §SA.4 partial closure $(1+A)\cdot(1-2A) = 0.9071$; E2 ZS-F11 §F11.1B J-fixed pivot $|5\rangle = J|5\rangle$; E3 ZS-F0 §11 F-BOOT-4 $J^2 = I$ two-traversals identity). [F] Anti-Numerology 2/2 PASS (F1 30.23341 vs. 30.7 relative difference 1.5% as OBSERVATION not identity; F2 independent derivation chains with no shared intermediate). [G] F-Gate Validation 1/1 PASS (G1 all six gates F-F13.1 through F-F13.6 formally specified with inputs, conditions, triggers, and present statuses). The Status: PLANNED qualifier of the immediately preceding paragraph is hereby retired and replaced operationally by Status: EXECUTED (2026-04-26). The seven-category structure of the registered checks, the "24 of 24 PASS, exit code 0" target stated below as a target-only at the time of initial release, and the description of the 50-digit mpmath precision regime are preserved verbatim in the original text above for historical record. No prior content of §10.1 is modified, removed, or weakened by this annotation. The execution outcome confirms the internal mathematical self-consistency of the v1.0 release across the seven registered categories at 50-digit mpmath precision and does not introduce any new physical claim, parameter, or prediction. The DERIVED-CONDITIONAL standing of Theorem F13.1 and Theorem F13.2B remains conditional on the same antecedents identified in the original release (ZS-A8 §6 Expansion-Contraction Symmetry status; ZS-M12 §6.2 $T_c \approx T_{reh}$ derivation). The anti-numerology Monte Carlo three-basket scan listed as PLANNED in the v1.0 Version History entry remains PLANNED and is not part of this execution annotation. Results JSON archive: zs_f13_results.json (machine-readable summary, 24/24 PASS).

Pre-registered in the Foundations theme alongside ZS-F0, ZS-F8, ZS-F9, ZS-F10, ZS-F11, ZS-F12. Closes ZS-A8 §SA.4 frame-metric component and the ZS-U8 §8 NC2 cycle-index ambiguity. Zero new free parameters; A = 35/437 remains the sole geometric input.

Acknowledgements & Code Availability

This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, structural analysis, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions.

The verification suite is registered as PLANNED. The verification script and Monte Carlo suite have not been executed at the time of this initial release; a future dated update to ZS-F13 v1.0 will report execution results.

Verification script (PLANNED): zs_f13_verify_v1_0.py
Dependencies: Python 3.10+, numpy, mpmath ≥ 50-digit precision, sympy
Execution (when implemented): python3 zs_f13_verify_v1_0.py
Target outcome (when implemented): 24 of 24 registered checks PASS, exit code 0 (target only, not yet executed)

The Monte Carlo anti-numerology suite zs_f13_mc_verify_v1_0.py (500,000-sample three-basket scan) is registered as PLANNED. Target output: H_NULL rejected at $p < 0.01\%$, STRONG PASS, exit code 0 (target, not yet executed).

Appendix A. Verification Suite Categories

The 24 registered checks are organized in seven categories. Implementation in zs_f13_verify_v1_0.py is PLANNED; targets are specified at machine precision or 50-digit mpmath precision as appropriate. Execution results will be reported in a future dated update to ZS-F13 v1.0.

Table A.1. Verification suite plan for zs_f13_verify_v1_0.py (PLANNED, not yet executed).

Cat.	Content	Registered Checks	Status
[A]	Locked Inputs	A = 35/437; Q = 11; (Z,X,Y) = (2,3,6); exp(A); $Y^2(1-2A)$ = 13212/437; exp(π/A)	5 checks, PLANNED
[B]	Theorem F13.1 numerics	$Y^2(1-2A)$ = 13212/437 ≈ 30.23341; (1−2A) = 367/437 ≈ 0.83982; conversion exp(π/A) ≈ 1.08×10^{17}; (1+A)(1−2A) = 0.9071	4 checks, PLANNED
[C]	Theorems F13.2A and F13.2B Z_2	V_E(+ε) − V_E(−ε) = 0 (algebraic, X-frame); n_s, r, η_B Z_2-symmetric numerical confirmation; mirror trajectory residual; F13.2B inheritance of F13.1 conditionality	4 checks, PLANNED

Cat.	Content	Registered Checks	Status
[D]	Möbius Chronology	ν entries Phase F to F' verified; n_strobo conversion 19.6 per Δν=1; Phase D τ_6/τ_5 = exp(π/A); R_Y/R_X ratio	5 checks, PLANNED
[E]	Cross-paper status	ZS-A8 §SA.4 partial closure verified; ZS-F11 §F11.1B link verified; ZS-F0 §11 F-BOOT-4 cosmological analog	3 checks, PLANNED
[F]	Anti-numerology	30.23341 vs 30.7 honest separation (no structural claim); independent derivation chains documented	2 checks, PLANNED
[G]	F-gate validation	All six gate conditions formally specified	1 check, PLANNED
TOTAL	—	—	24 registered checks, all PLANNED

A.2 Honest Limitations

(L1) Theorem F13.1 is DERIVED-CONDITIONAL with explicit conditionality registered in NC-F13.4 (Y-observation-of-X is OPEN) and F-F13.5 (Y-internal channel-capacity computation pending). The conditionality further inherits the ZS-A8 §6 Theorem 6.1 conditionality on dim(Z) = 2 (itself five-fold over-determined per ZS-F0 v1.0(R) Corollary 5.2.A.2). The status is NOT "operationally DERIVED" in the unqualified sense; it is DERIVED-CONDITIONAL.

(L2) The Y-internal frame computation of Y-characteristic radius $R_Y = \ell_P \times Y^2(1-2A) = \ell_P \times 13212/437$ is structural under ZS-A8 §6 Theorem 6.1, but a fully independent Y-internal channel-capacity calculation analogous to ZS-Q7 §6 X-internal calculation has not been performed. F-F13.5 records this as OPEN.

(L3) The Y-observation-of-X direction (NC-F13.4) is OPEN. The parallel reading suggests no dilation, but full closure requires Y-internal channel-capacity work.

(L4) The 1.5% T_c / T_{reh} gap (F-F13.3) is OPEN at the structural-explanation level. The numerical agreement is documented in ZS-M12 §6.2; a closed-form derivation showing $T_c = T_{reh}$ exactly under additional structural assumptions would close F-F13.3.

(L5) Theorem F13.2A (DERIVED) and Theorem F13.2B (DERIVED-CONDITIONAL on F13.1) establish operational unobservability within Z-Spin. Neither addresses whether a meta-theory containing Z-Spin could provide a cycle-index access mechanism. NC-F13.2 retains this scope boundary.

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Version History

v1.0 (April 2026): Initial public release. Theorem F13.1 (Y-Observer Frame Scale Closure, DERIVED-CONDITIONAL): explicit Y-frame operational scale trio (T_Y, R_Y, η_Y) = (τ_5 in Y-clock, $\ell_P \times 13212/437$, $367/437$) constructed as the structural counterpart to the X-frame trio (T_X, R_X, η_X) = (t, c/H_0, $\exp(A)$) under ZS-A8 §6 Expansion-Contraction Symmetry. The term "frame scale trio" denotes the operational triple (clock proper time, characteristic radius, multiplicative scale-flow factor); §3.0 terminology note distinguishes this from a derived spacetime metric tensor. Theorem F13.2A (X-Frame Cycle Index Unobservability, DERIVED) and Theorem F13.2B (Y-Frame Extension, DERIVED-CONDITIONAL on F13.1): the cumulative cycle index k is structurally unobservable in the X-frame from PROVEN/DERIVED inputs alone ($V_E(-\varepsilon) \equiv V_E(+\varepsilon)$, Z_2 symmetry of observables, Auto-Surgery reset), and extends to the Y-frame conditional on F13.1. Möbius identification realized operationally at $T_c \approx T_{reh} \approx 2.48\times10^{15}$ GeV (DERIVED, ZS-M12 §6.2). Unified Möbius Chronology Table 4 consolidates seven principal phases F→A→B→C→D→E→F' plus three transition subphases (Reheating, E', E") in three time coordinates and three space-frame scales. Six falsification gates F-F13.1 through F-F13.6 pre-registered: F-F13.1 PASS (ZS-A8 §6 stable), F-F13.2 PASS (PROVEN identity), F-F13.3 OPEN (1.5% $T_c - T_{reh}$ gap), F-F13.4 TESTABLE / NOT TRIGGERED (no cycle-distinguishing observation reported), F-F13.5 OPEN (Y-internal channel-capacity pending), F-F13.6 OPEN (anti-overclaim guard). Six non-claims NC-F13.1 through NC-F13.6 explicit (including

NC-F13.4 Y-observation-of-X OPEN, NC-F13.5 Möbius is structural not phenomenological, NC-F13.6 30.23341 vs 30.7 honest separation). Anti-numerology Monte Carlo three-basket scan PLANNED. Verification Plan: 24 registered checks; implementation pending in zs_f13_verify_v1_0.py at 50-digit mpmath precision; execution results to be reported in future dated update. Zero new free parameters; all inputs LOCKED, PROVEN, or DERIVED in prior corpus papers (29 upstream dependencies tracked in §2 Table 1). Pre-registered in Foundations theme alongside ZS-F0, ZS-F8, ZS-F9, ZS-F10, ZS-F11, ZS-F12. Closes ZS-A8 v1.0 R §SA.4 frame-scale component (partial closure of HYPOTHESIS-strong INTERPRETATION) and ZS-U8 v1.0 §8 NC2 cycle-index ambiguity. All numerical values use exact rational forms where available ($1-2A = 367/437 = 0.83981693...$, $Y^2(1-2A) = 13212/437 = 30.23341...$ per ZS-A8 §4.3 Conformal Form table). Word count strictly increased relative to all referenced upstream papers per the v1.0 freeze convention. (Consolidated from internal Z-Spin Collaboration research notes through April 2026.)

v1.0 — 2026-04-26 update (verification execution): The 24-check verification suite zs_f13_verify_v1_0.py was executed on 2026-04-26 at 50-digit mpmath precision. Result: TOTAL 24/24 PASS, exit code 0. The Verification Summary header on the title page is updated in place from "Verification Plan: 24 registered checks | Implementation Pending | Zero Free Parameters | Zero New Physical Predictions" to "Verification: 24/24 PASS (executed 2026-04-26) | Zero Free Parameters | Zero New Physical Predictions", mirroring the live-status-card convention used in ZS-S2 v1.0, ZS-S5 v1.0, and other corpus papers whose verification suites have been executed. The §10.1 Verification Plan body paragraph is preserved verbatim, and a dated "Execution Annotation (2026-04-26 dated update, in-place per Z-Spin v1.0 freeze convention)" paragraph is appended immediately after it, reporting per-category outcomes for [A] Locked Inputs 5/5 PASS, [B] Theorem F13.1 Numerics 4/4 PASS, [C] Theorems F13.2A and F13.2B Z_2-Symmetry 4/4 PASS, [D] Möbius Chronology 5/5 PASS, [E] Cross-Paper Status 3/3 PASS, [F] Anti-Numerology 2/2 PASS, and [G] F-Gate Validation 1/1 PASS. The three downstream §10.1 sub-paragraphs that describe (i) the seven-category structure of the registered checks, (ii) the "Target outcome (when implemented): 24 of 24 registered checks PASS, exit code 0 (target only, not yet executed)" line, and (iii) the §10.2 cross-reference matrix entry stating "24 registered checks, all PLANNED" are preserved verbatim for historical record; the new Execution Annotation paragraph is the single point of operational status update. The Abstract paragraph mentioning "Verification Plan: 24 registered checks; implementation pending in zs_f13_verify_v1_0.py at 50-digit mpmath precision" is likewise preserved verbatim for historical record. No theorem statement, no falsification gate status, no NC clause, no Table 1/4/5/6 entry, and no derivation chain is modified by this dated update; the execution outcome confirms internal mathematical self-consistency of the v1.0 release across the seven registered categories at 50-digit mpmath precision and does not introduce any new physical claim, parameter, or prediction. The DERIVED-CONDITIONAL standing of Theorem F13.1 (Y-Observer Frame Scale Closure) and Theorem F13.2B (Y-Frame Cycle Index Unobservability extension) remains conditional on the same antecedents identified in the original release: the ZS-A8 §6 Expansion-Contraction Symmetry status (HYPOTHESIS-strong INTERPRETATION; partial closure carried through $(1+A)(1-2A) = 0.9071$ in test E1) and the ZS-M12 §6.2 $T_c \approx T_{reh} \approx 2.48\times10^{15}$ GeV Möbius identification (DERIVED). The PASS standing of F-F13.1 (ZS-A8 §6 stable) and F-F13.2 (PROVEN identity) is reconfirmed at 50-digit mpmath precision; the OPEN standings of F-F13.3 (1.5% $T_c - T_{reh}$ gap), F-F13.5 (Y-internal channel-capacity pending), and F-F13.6 (anti-overclaim guard), and the TESTABLE / NOT TRIGGERED standing of F-F13.4 (no cycle-distinguishing observation reported) are unchanged by the present

execution. The 30.23341 vs. 30.7 anti-numerology honest-separation registered as NC-F13.6 is reconfirmed (test F1 PASS at 1.5% relative difference, OBSERVATION not identity); the F2 independent-derivation-chains test confirms that the $(1-2A) = 367/437$ conformal factor of ZS-A8 §4.3 and the $W^2 = I$ conjugation involution of ZS-F11 §F11.1B (J-fixed pivot $|5\rangle = J|5\rangle$) and the ZS-F0 §11 F-BOOT-4 ($J^2 = I$ two-traversals identity) share no intermediate quantity in their derivation paths, satisfying the anti-numerology criterion of independent chains. The anti-numerology Monte Carlo three-basket scan listed as PLANNED in the v1.0 Version History entry remains PLANNED and is not in the scope of the present execution. Word count strictly increased relative to the v1.0 release per the v1.0 freeze convention; all numerical values remain in exact rational form where available ($1-2A = 367/437 = 0.83981693...$, $Y^2(1-2A) = 13212/437 = 30.23341...$, per ZS-A8 §4.3 Conformal Form table). External label remains v1.0 (no version bump, no citation cascade across the corpus). Zero new free parameters; $A = 35/437$ remains the sole geometric input. Results JSON archive: zs_f13_results.json (machine-readable per-test summary, 24/24 PASS, generated by zs_f13_verify_v1_0.py at 50-digit mpmath precision). This dated update completes the verification-execution transition for ZS-F13 v1.0, moving the paper from "Plan registered, implementation pending" to "Plan registered, implementation executed, 24/24 PASS" within the Z-Spin v1.0 freeze convention; the seven-category coverage (Locked Inputs / Theorem F13.1 numerics / Z_2-symmetry of F13.2A and F13.2B / Möbius Chronology / Cross-paper status / Anti-numerology / F-gate validation) is fully exercised, and the present dated entry serves as the single record of the execution outcome.